Day 133 | Unit 09 | Teacher plan

Display, Describe, and Summarize Data

Compare Data Distributions

Standards: 6.SP.A.2, 6.SP.B.5d

Learning target: I can compare centers and variability in two data sets.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Compute comparable measures for both groups. Compare typical values and consistency using the same units. State what the data support without claiming that every member of one group exceeds every member of the other.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 10

Day 133 | Unit 09 | Lesson

Display, Describe, and Summarize Data

Compare Data Distributions

Standards: 6.SP.A.2, 6.SP.B.5d

Learning target: I can compare centers and variability in two data sets.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

01 / Learn

Compute comparable measures for both groups. Compare typical values and consistency using the same units. State what the data support without claiming that every member of one group exceeds every member of the other.

Worked example 1

Reading times (min): Group A [6, 7, 7, 8, 8, 9], Group B [8, 9, 9, 10, 10, 11]. Compare median and IQR. Which group reads longer typically? Which is more consistent?

1. Compute comparable measures for both groups.

2. Compare typical values and consistency using the same units.

3. Adding 2 shifts all values and the median by 2, but does not change Q3 - Q1.

4. State the result in the form requested: A median 15/2, B median 19/2 min; both IQR 1 min. B is typically 2 min higher; equal middle-half variability..

Answer: A median 15/2, B median 19/2 min; both IQR 1 min. B is typically 2 min higher; equal middle-half variability.

Worked example 2

Six students recorded reading times (minutes): 2, 4, 4, 6, 6, 8. A second group has every corresponding time 5 minutes larger. Part A: compare medians. Part B: compare IQRs and justify without listing all new data.

1. Compute comparable measures for both groups.

2. Adding 5 shifts every position and both quartiles by 5.

3. The median increases by 5

4. Q3 - Q1 remains 2.

Answer: First median 5; second 10 min; both IQR 2 min.

Day 133 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Reading times (min): Group A [8, 9, 9, 10, 10, 11], Group B [10, 11, 11, 12, 12, 13]. Compare median and IQR. Which group reads longer typically? Which is more consistent?

02 / Constructed response

Reading times (min): Group A [4, 5, 5, 6, 6, 7], Group B [6, 7, 7, 8, 8, 9]. Compare median and IQR. Which group reads longer typically? Which is more consistent?

03 / Constructed response

Reading times (min): Group A [7, 8, 8, 9, 9, 10], Group B [9, 10, 10, 11, 11, 12]. Compare median and IQR. Which group reads longer typically? Which is more consistent?

04 / Constructed response

Reading times (min): Group A [8, 12, 12, 16, 16, 20], Group B [10, 14, 14, 18, 18, 22]. Compare median and IQR. Which group reads longer typically? Which is more consistent?

05 / Constructed response

Reading times (min): Group A [6, 9, 9, 12, 12, 15], Group B [8, 11, 11, 14, 14, 17]. Compare median and IQR. Which group reads longer typically? Which is more consistent?

06 / Constructed response

Reading times (min): Group A [6, 8, 8, 10, 10, 12], Group B [8, 10, 10, 12, 12, 14]. Compare median and IQR. Which group reads longer typically? Which is more consistent?

07 / Constructed response

Reading times (min): Group A [4, 5, 5, 6, 7, 16], Group B [6, 7, 7, 8, 9, 18]. Compare median and IQR. Which group reads longer typically? Which is more consistent?

08 / Two-part comparison and reasoning

Six students recorded reading times (minutes): 5, 10, 10, 15, 15, 20. A second group has every corresponding time 4 minutes larger. Part A: compare medians. Part B: compare IQRs and justify without listing all new data.

09 / Two-part comparison and reasoning

Six students recorded reading times (minutes): 2, 5, 5, 8, 8, 11. A second group has every corresponding time 2 minutes larger. Part A: compare medians. Part B: compare IQRs and justify without listing all new data.

10 / Constructed response

Reading times (min): Group A [6, 7, 7, 8, 9, 18], Group B [8, 9, 9, 10, 11, 20]. Compare median and IQR. Which group reads longer typically? Which is more consistent?

Day 133 | Unit 09 | Answer key

Display, Describe, and Summarize Data

Compare Data Distributions

Standards: 6.SP.A.2, 6.SP.B.5d

Learning target: I can compare centers and variability in two data sets.

1. A median 19/2, B median 23/2 min; both IQR 1 min. B is typically 2 min higher; equal middle-half variability.

Adding 2 shifts all values and the median by 2, but does not change Q3 - Q1.

6.SP.A.2, 6.SP.B.5d | Constructed response

2. A median 11/2, B median 15/2 min; both IQR 1 min. B is typically 2 min higher; equal middle-half variability.

Adding 2 shifts all values and the median by 2, but does not change Q3 - Q1.

6.SP.A.2, 6.SP.B.5d | Constructed response

3. A median 17/2, B median 21/2 min; both IQR 1 min. B is typically 2 min higher; equal middle-half variability.

Adding 2 shifts all values and the median by 2, but does not change Q3 - Q1.

6.SP.A.2, 6.SP.B.5d | Constructed response

4. A median 14, B median 16 min; both IQR 4 min. B is typically 2 min higher; equal middle-half variability.

Adding 2 shifts all values and the median by 2, but does not change Q3 - Q1.

6.SP.A.2, 6.SP.B.5d | Constructed response

5. A median 21/2, B median 25/2 min; both IQR 3 min. B is typically 2 min higher; equal middle-half variability.

Adding 2 shifts all values and the median by 2, but does not change Q3 - Q1.

6.SP.A.2, 6.SP.B.5d | Constructed response

6. A median 9, B median 11 min; both IQR 2 min. B is typically 2 min higher; equal middle-half variability.

Adding 2 shifts all values and the median by 2, but does not change Q3 - Q1.

6.SP.A.2, 6.SP.B.5d | Constructed response

7. A median 11/2, B median 15/2 min; both IQR 2 min. B is typically 2 min higher; equal middle-half variability.

Adding 2 shifts all values and the median by 2, but does not change Q3 - Q1.

6.SP.A.2, 6.SP.B.5d | Constructed response

8. First median 25/2; second 33/2 min; both IQR 5 min.

Adding 4 shifts every position and both quartiles by 4. The median increases by 4; Q3 - Q1 remains 5.

6.SP.A.2, 6.SP.B.5d | Two-part comparison and reasoning

9. First median 13/2; second 17/2 min; both IQR 3 min.

Adding 2 shifts every position and both quartiles by 2. The median increases by 2; Q3 - Q1 remains 3.

6.SP.A.2, 6.SP.B.5d | Two-part comparison and reasoning

10. A median 15/2, B median 19/2 min; both IQR 2 min. B is typically 2 min higher; equal middle-half variability.

Adding 2 shifts all values and the median by 2, but does not change Q3 - Q1.

6.SP.A.2, 6.SP.B.5d | Constructed response